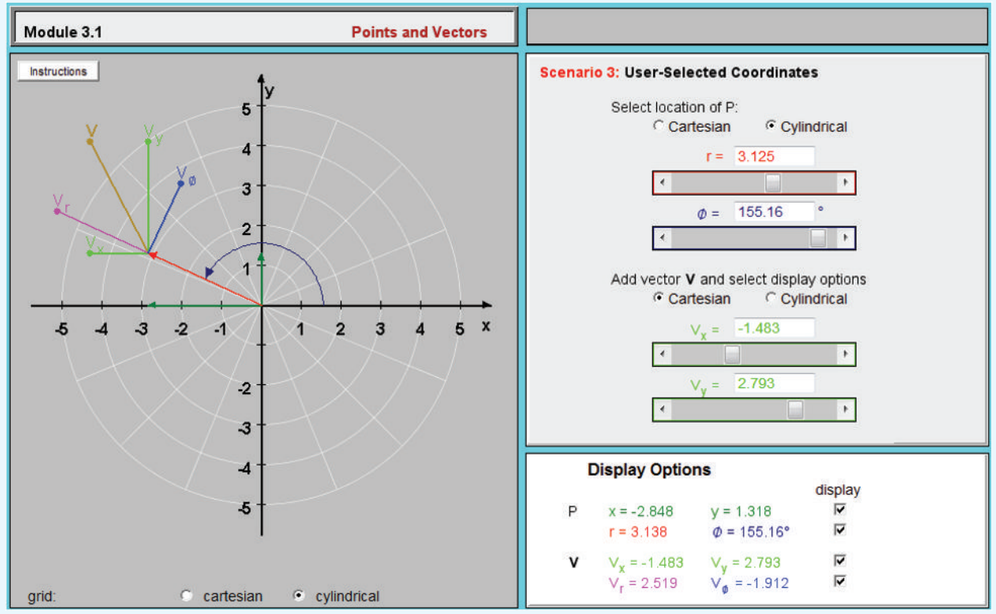
Module 3.1 Points and Vectors Examine the relationships between Cartesian coordinates (x, y) and cylindrical coordinates (r, ϕ) for points and vectors.



Solution: The prescribed surface is shown in Fig. 3-12. Use of Eq. (3.43a) for a surface element with constant r gives

$$S = r\int_{\phi=30^\circ}^{60^\circ} d\phi \int_{z=0}^{3} dz = 5\phi\Big|_{\pi/6}^{\pi/3} z\Big|_0^3 = \frac{5\pi}{2}.$$

Note that ϕ had to be converted to radians before evaluating the integration limits.

Exercise 3-6: A circular cylinder of radius $r = 5$ cm is concentric with the z axis and extends between $z = -3$ cm and $z = 3$ cm. Use Eq. (3.44) to find the cylinder's volume.

Answer: 471.2 cm^3. (See EM.)

3-2.3 Spherical Coordinates

In the spherical coordinate system, the location of a point in space is uniquely specified by the variables R, θ, and ϕ (Fig. 3-13). The range coordinate R, which measures the distance from the origin to the point, describes a sphere of radius R centered at the origin. The *zenith angle* θ is measured from the positive z axis and it describes a conical surface with its apex at the origin, and the azimuth angle ϕ is the same as in cylindrical coordinates. The ranges of R, θ, and ϕ are $0 \le R < \infty$, $0 \le \theta \le \pi$, and $0 \le \phi < 2\pi$. The base vectors $\hat{\mathbf{R}}$, $\hat{\boldsymbol{\theta}}$, and $\hat{\boldsymbol{\phi}}$ obey the right-hand cyclic relations:

$$\hat{\mathbf{R}} \times \hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\phi}}, \quad \hat{\boldsymbol{\theta}} \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{R}}, \quad \hat{\boldsymbol{\phi}} \times \hat{\mathbf{R}} = \hat{\boldsymbol{\theta}}. \tag{3.45}$$

A vector with components A_R, A_θ, and A_ϕ is written as

$$\mathbf{A} = \hat{\mathbf{a}}|\mathbf{A}| = \hat{\mathbf{R}}A_R + \hat{\boldsymbol{\theta}}A_\theta + \hat{\boldsymbol{\phi}}A_\phi, \tag{3.46}$$

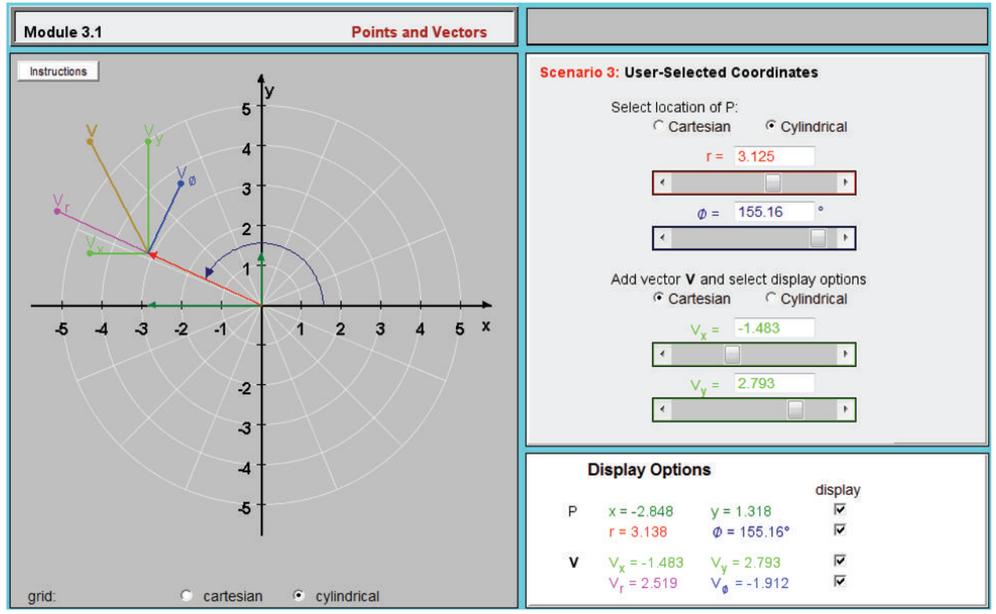
and its magnitude is

$$|\mathbf{A}| = \sqrt[+]{\mathbf{A}\cdot\mathbf{A}} = \sqrt[+]{A_R^2 + A_\theta^2 + A_\phi^2}. \tag{3.47}$$

The position vector of point $P(R_1, \theta_1, \phi_1)$ is simply

$$\mathbf{R}_1 = \overrightarrow{OP} = \hat{\mathbf{R}}R_1, \tag{3.48}$$

模块 3.1 点和向量 检查点和向量的笛卡尔坐标 (x, y) 和柱坐标 (r, ϕ) 之间的关系。



解：规定曲面如图3-12所示。使用等式。 (3.43a) 对于具有常数 r 的表面元素，给出

$$S = r\int_{\phi=30^\circ}^{60^\circ} d\phi \int_{z=0}^{3} dz = 5\phi\Big|_{\pi/6}^{\pi/3} z\Big|_0^3 = \frac{5\pi}{2}.$$

请注意，在评估积分限制之前，必须将 ϕ 转换为弧度。

练习 3-6：半径为 r = 5 cm 的圆柱体与 z 轴同心，并在 z = −3 cm 和 z = 3 cm 之间延伸。使用方程式。 (3.44) 求圆柱体的体积。

答案：471.2 厘米^3。 （参见EM。）

3-2.3 球坐标

在球坐标系中，空间中点的位置由变量 R、θ 和 ϕ (唯一指定。 3-13)。范围坐标 R，测量从原点到该点的距离，描述了以原点为中心、半径为 R 的球体。天顶角 θ 是从 z 轴正方向测量的，它描述了一个以原点为顶点的圆锥面，而方位角 ϕ 与柱坐标系中的相同。 R、θ和ϕ的范围是0≤R< ∞、0≤ θ ≤ π和0≤ ϕ <2 π。基向量 R̂、$\hat{\boldsymbol{\theta}}$ 和 $\hat{\boldsymbol{\phi}}$ 服从右手循环关系：

$$\hat{\mathbf{R}} \times \hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\phi}}, \quad \hat{\boldsymbol{\theta}} \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{R}}, \quad \hat{\boldsymbol{\phi}} \times \hat{\mathbf{R}} = \hat{\boldsymbol{\theta}}. \tag{3.45}$$

具有分量 A_R、A_θ 和 A_ϕ 的向量可写为

$$\mathbf{A} = \hat{\mathbf{a}}|\mathbf{A}| = \hat{\mathbf{R}}A_R + \hat{\boldsymbol{\theta}}A_\theta + \hat{\boldsymbol{\phi}}A_\phi, \tag{3.46}$$

它的大小是

$$|\mathbf{A}| = \sqrt[+]{\mathbf{A}\cdot\mathbf{A}} = \sqrt[+]{A_R^2 + A_\theta^2 + A_\phi^2}. \tag{3.47}$$

点 $P(R_1, \theta_1, \phi_1)$ 的位置向量就是

$$\mathbf{R}_1 = \overrightarrow{OP} = \hat{\mathbf{R}}R_1, \tag{3.48}$$

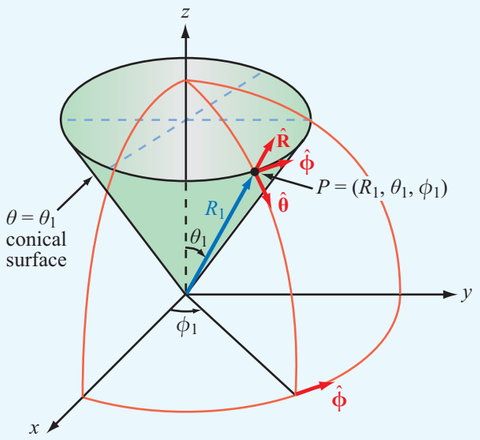


Figure 3-13 Point $P(R_1, \theta_1, \phi_1)$ in spherical coordinates.

As shown in **Fig. 3-14**, the differential lengths along $\hat{\mathbf{R}}$, $\hat{\boldsymbol{\theta}}$, and $\hat{\boldsymbol{\phi}}$ are

$$dl_R = dR, \quad dl_\theta = R\,d\theta, \quad dl_\phi = R\sin\theta\,d\phi. \tag{3.49}$$

Hence, the expressions for the vector differential length $d\mathbf{l}$, the vector differential surface $d\mathbf{s}$, and the differential volume $d\mathcal{V}$ are

$$\begin{aligned} d\mathbf{l} &= \hat{\mathbf{R}}\,dl_R + \hat{\boldsymbol{\theta}}\,dl_\theta + \hat{\boldsymbol{\phi}}\,dl_\phi \\ &= \hat{\mathbf{R}}\,dR + \hat{\boldsymbol{\theta}}R\,d\theta + \hat{\boldsymbol{\phi}}R\sin\theta\,d\phi, \end{aligned} \tag{3.50a}$$

$$d\mathbf{s}_R = \hat{\mathbf{R}}\,dl_\theta\,dl_\phi = \hat{\mathbf{R}}R^2\sin\theta\,d\theta\,d\phi \tag{3.50b}$$

(θ–ϕ spherical surface),

$$d\mathbf{s}_\theta = \hat{\boldsymbol{\theta}}\,dl_R\,dl_\phi = \hat{\boldsymbol{\theta}}R\sin\theta\,dR\,d\phi \tag{3.50c}$$

(R–ϕ conical surface),

$$d\mathbf{s}_\phi = \hat{\boldsymbol{\phi}}\,dl_R\,dl_\theta = \hat{\boldsymbol{\phi}}R\,dR\,d\theta \quad (R\text{–}\theta \text{ plane}), \tag{3.50d}$$

$$d\mathcal{V} = dl_R\,dl_\theta\,dl_\phi = R^2\sin\theta\,dR\,d\theta\,d\phi. \tag{3.50e}$$

These relations are summarized in **Table 3-1**.

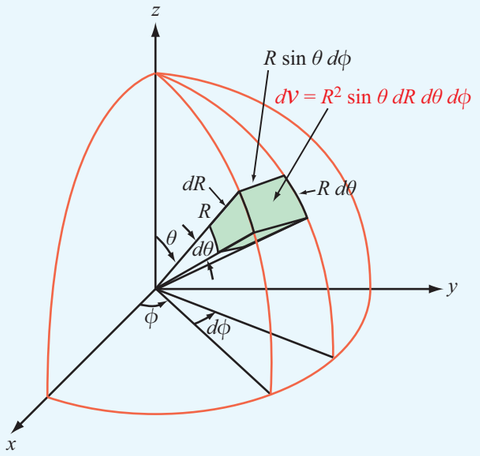


Figure 3-14 Differential volume in spherical coordinates.

while keeping in mind that $\hat{\mathbf{R}}$ is implicitly dependent on θ_1 and ϕ_1.

Example 3-6: Surface Area in Spherical Coordinates

The spherical strip shown in **Fig. 3-15** is a section of a sphere of radius 3 cm. Find the area of the strip.

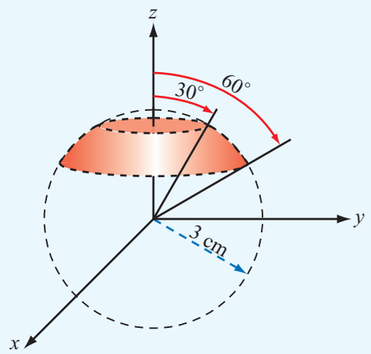


Figure 3-15 Spherical strip of Example 3-6.

142 CHAPTER 3 VECTOR ANALYSIS

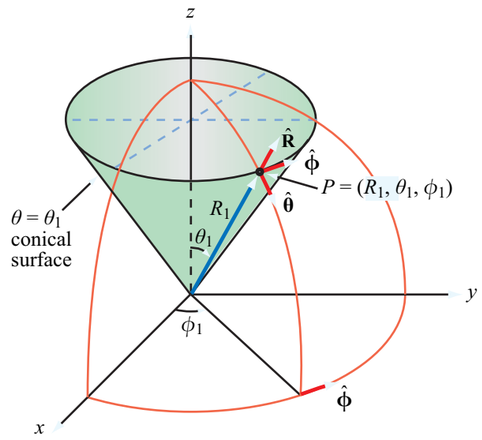


图 3-13 球坐标中的点 $P(R_1,\theta_1,\phi_1)$。

如图3-14所示，沿R̂、θ̂和φ̂的差分长度为

$$dl_R = dR, \quad dl_\theta = R\,d\theta, \quad dl_\phi = R\sin\theta\,d\phi. \tag{3.49}$$

因此，矢量微分长度 dl、矢量微分面 ds 和微分体积 dυ 的表达式为

$$\begin{aligned} d\mathbf{l} &= \hat{\mathbf{R}}\,dl_R + \hat{\boldsymbol{\theta}}\,dl_\theta + \hat{\boldsymbol{\phi}}\,dl_\phi \\ &= \hat{\mathbf{R}}\,dR + \hat{\boldsymbol{\theta}}R\,d\theta + \hat{\boldsymbol{\phi}}R\sin\theta\,d\phi, \end{aligned} \tag{3.50a}$$

$$d\mathbf{s}_R = \hat{\mathbf{R}}\,dl_\theta\,dl_\phi = \hat{\mathbf{R}}R^2\sin\theta\,d\theta\,d\phi \tag{3.50b}$$

(θ–ϕ spherical surface),

$$d\mathbf{s}_\theta = \hat{\boldsymbol{\theta}}\,dl_R\,dl_\phi = \hat{\boldsymbol{\theta}}R\sin\theta\,dR\,d\phi \tag{3.50c}$$

(R–ϕ conical surface),

$$d\mathbf{s}_\phi = \hat{\boldsymbol{\phi}}\,dl_R\,dl_\theta = \hat{\boldsymbol{\phi}}R\,dR\,d\theta \quad (R\text{–}\theta \text{ plane}), \tag{3.50d}$$

$$d\mathcal{V} = dl_R\,dl_\theta\,dl_\phi = R^2\sin\theta\,dR\,d\theta\,d\phi. \tag{3.50e}$$

表 3-1 总结了这些关系。

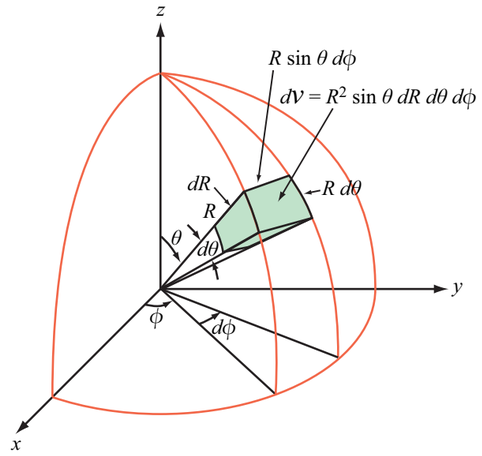


图 3-14 球坐标中的微分体积。

同时请记住，R̂ 隐式依赖于 θ_1 和 ϕ_1。

示例 3-6：球坐标中的表面积

图3-15所示的球形条是半径为3厘米的球体的截面。求该条带的面积。

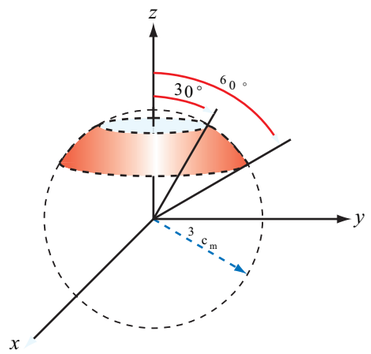


图3-15 实施例3-6的球形条。

Solution: Use of Eq. (3.50b) for the area of an elemental spherical area with constant radius R gives

$$\begin{aligned} S &= R^2 \int_{\theta=30^\circ}^{60^\circ} \sin\theta \, d\theta \int_{\phi=0}^{2\pi} d\phi \\ &= 9(-\cos\theta)\Big|_{30^\circ}^{60^\circ} \phi\Big|_0^{2\pi} \qquad (\text{cm}^2) \\ &= 18\pi(\cos 30^\circ - \cos 60^\circ) = 20.7 \text{ cm}^2. \end{aligned}$$

Example 3-7: Charge in a Sphere

A sphere of radius 2 cm contains a volume charge density ρ_v given by

$$\rho_v = 4\cos^2\theta \qquad (\text{C/m}^3).$$

Find the total charge Q contained in the sphere.

Solution:

$$\begin{aligned} Q &= \int_{\upsilon} \rho_v \, d\upsilon \\ &= \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \int_{R=0}^{2\times10^{-2}} (4\cos^2\theta) R^2 \sin\theta \, dR \, d\theta \, d\phi \\ &= 4 \int_0^{2\pi} \int_0^{\pi} \left(\frac{R^3}{3}\right)\Big|_0^{2\times10^{-2}} \sin\theta\cos^2\theta \, d\theta \, d\phi \\ &= \frac{32}{3} \times 10^{-6} \int_0^{2\pi} \left(-\frac{\cos^3\theta}{3}\right)\Big|_0^{\pi} d\phi \\ &= \frac{64}{9} \times 10^{-6} \int_0^{2\pi} d\phi \\ &= \frac{128\pi}{9} \times 10^{-6} = 44.68 \qquad (\mu\text{C}). \end{aligned}$$

Note that the limits on R were converted to meters prior to evaluating the integral on R.

3-3 Transformations between Coordinate Systems

The position of a given point in space of course does not depend on the choice of coordinate system. That is, its location is the same irrespective of which specific coordinate system is used to represent it. The same is true for vectors. Nevertheless, certain coordinate systems may be more useful than others in solving a given problem, so it is essential that we have the tools to "translate" the problem from one system to another. In this section, we shall establish the relations between the variables (x,y,z) of the Cartesian system, (r,ϕ,z) of the cylindrical system, and (R,θ,ϕ) of the spherical system. These relations will then be used to transform expressions for vectors expressed in any one of the three systems into expressions applicable in the other two.

3-3.1 Cartesian to Cylindrical Transformations

Point P in Fig. 3-16 has Cartesian coordinates (x,y,z) and cylindrical coordinates (r,ϕ,z). Both systems share the coordinate z, and the relations between the other two pairs of coordinates can be obtained from the geometry in Fig. 3-16. They are

$$r = \sqrt[+]{x^2+y^2}, \qquad \phi = \tan^{-1}\left(\frac{y}{x}\right), \tag{3.51}$$

and the inverse relations are

$$x = r\cos\phi, \qquad y = r\sin\phi. \tag{3.52}$$

Next, with the help of Fig. 3-17, which shows the directions of the unit vectors $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, $\hat{\mathbf{r}}$, and $\hat{\boldsymbol{\phi}}$ in the x–y plane, we obtain the relations:

$$\hat{\mathbf{r}}\cdot\hat{\mathbf{x}} = \cos\phi, \qquad \hat{\mathbf{r}}\cdot\hat{\mathbf{y}} = \sin\phi, \tag{3.53a}$$

$$\hat{\boldsymbol{\phi}}\cdot\hat{\mathbf{x}} = -\sin\phi, \qquad \hat{\boldsymbol{\phi}}\cdot\hat{\mathbf{y}} = \cos\phi. \tag{3.53b}$$

To express $\hat{\mathbf{r}}$ in terms of $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$, we write $\hat{\mathbf{r}}$ as

$$\hat{\mathbf{r}} = \hat{\mathbf{x}}a + \hat{\mathbf{y}}b, \tag{3.54}$$

where a and b are unknown transformation coefficients. The dot product $\hat{\mathbf{r}}\cdot\hat{\mathbf{x}}$ gives

$$\hat{\mathbf{r}}\cdot\hat{\mathbf{x}} = \hat{\mathbf{x}}\cdot\hat{\mathbf{x}}a + \hat{\mathbf{y}}\cdot\hat{\mathbf{x}}b = a. \tag{3.55}$$

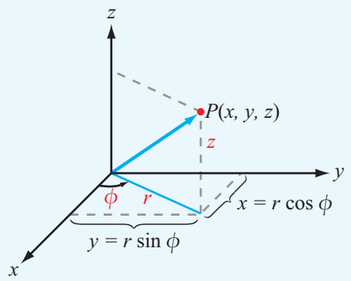


Figure 3-16 Interrelationships between Cartesian coordinates (x,y,z) and cylindrical coordinates (r,ϕ,z).

解决方案：使用方程。 (3.50b) 对于具有恒定半径 R 的基本球形区域的面积，给出

$$\begin{aligned} S &= R^2 \int_{\theta=30^\circ}^{60^\circ} \sin\theta \, d\theta \int_{\phi=0}^{2\pi} d\phi \\ &= 9(-\cos\theta)\Big|_{30^\circ}^{60^\circ} \phi\Big|_0^{2\pi} \qquad (\text{cm}^2) \\ &= 18\pi(\cos 30^\circ - \cos 60^\circ) = 20.7 \text{ cm}^2. \end{aligned}$$

例 3-7：球体中的电荷

半径为 2 cm 的球体包含的体积电荷密度 ρ_v 为

$$\rho_v = 4\cos^2\theta \qquad (\text{C/m}^3).$$

求球体中所含的总电荷 Q。

解决方案：

$$\begin{aligned} Q &= \int_{\upsilon} \rho_v \, d\upsilon \\ &= \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \int_{R=0}^{2\times10^{-2}} (4\cos^2\theta) R^2 \sin\theta \, dR \, d\theta \, d\phi \\ &= 4 \int_0^{2\pi} \int_0^{\pi} \left(\frac{R^3}{3}\right)\Big|_0^{2\times10^{-2}} \sin\theta\cos^2\theta \, d\theta \, d\phi \\ &= \frac{32}{3} \times 10^{-6} \int_0^{2\pi} \left(-\frac{\cos^3\theta}{3}\right)\Big|_0^{\pi} d\phi \\ &= \frac{64}{9} \times 10^{-6} \int_0^{2\pi} d\phi \\ &= \frac{128\pi}{9} \times 10^{-6} = 44.68 \qquad (\mu\text{C}). \end{aligned}$$

请注意，在评估 R 的积分之前，R 的限值已转换为米。

3-3 坐标系之间的变换

当然，空间中给定点的位置不取决于坐标系的选择。也就是说，无论使用哪种特定坐标系来表示它，它的位置都是相同的。对于向量也是如此。然而，在解决给定问题时，某些坐标系可能比其他坐标系更有用，因此我们拥有将问题从一个系统“转换”到另一个系统的工具至关重要。在这个

在本节中，我们将建立笛卡尔系统的变量 (x, y, z)、圆柱系统的变量 (r, ϕ, z) 和球面系统的变量 (R, θ, ϕ) 之间的关系。然后，这些关系将用于将三个系统中任何一个系统中表达的向量表达式转换为适用于其他两个系统的表达式。

3-3.1 笛卡尔到圆柱的变换

图3-16中的点P有笛卡尔坐标（x，y，z）和柱坐标（r，ϕ，z）。两个系统共享坐标z，另外两对坐标之间的关系可以从图3-16的几何结构中得到。他们是

$$r = \sqrt[+]{x^2+y^2}, \qquad \phi = \tan^{-1}\left(\frac{y}{x}\right), \tag{3.51}$$

反比关系是

$$x = r\cos\phi, \qquad y = r\sin\phi. \tag{3.52}$$

接下来，借助图 3-17（显示 x-y 平面中单位向量 x̂、ŷ、r̂ 和 $\hat{\boldsymbol{\phi}}$ 的方向），我们得到以下关系：

$$\hat{\mathbf{r}}\cdot\hat{\mathbf{x}} = \cos\phi, \qquad \hat{\mathbf{r}}\cdot\hat{\mathbf{y}} = \sin\phi, \tag{3.53a}$$

$$\hat{\boldsymbol{\phi}}\cdot\hat{\mathbf{x}} = -\sin\phi, \qquad \hat{\boldsymbol{\phi}}\cdot\hat{\mathbf{y}} = \cos\phi. \tag{3.53b}$$

为了用 x̂ 和 ŷ 表示 r̂，我们将 r̂ 写为

$$\hat{\mathbf{r}} = \hat{\mathbf{x}}a + \hat{\mathbf{y}}b, \tag{3.54}$$

其中 a 和 b 是未知的变换系数。点积 r̂ · x̂ 给出

$$\hat{\mathbf{r}}\cdot\hat{\mathbf{x}} = \hat{\mathbf{x}}\cdot\hat{\mathbf{x}}a + \hat{\mathbf{y}}\cdot\hat{\mathbf{x}}b = a. \tag{3.55}$$

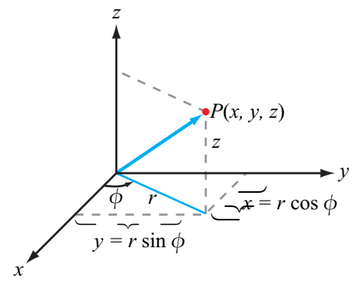


图 3-16 笛卡尔坐标 (x,y,z) 和柱坐标 (r,ϕ,z) 之间的相互关系。

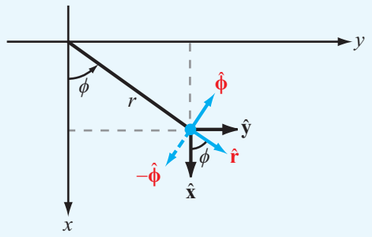


Figure 3-17 Interrelationships between base vectors $(\hat{\mathbf{x}},\hat{\mathbf{y}})$ and $(\hat{\mathbf{r}},\hat{\boldsymbol{\phi}})$.

Comparison of Eq. (3.55) with Eq. (3.53a) yields $a = \cos\phi$. Similarly, application of the dot product $\hat{\mathbf{r}}\cdot\hat{\mathbf{y}}$ to Eq. (3.54) gives $b = \sin\phi$. Hence,

$$\hat{\mathbf{r}} = \hat{\mathbf{x}}\cos\phi + \hat{\mathbf{y}}\sin\phi. \tag{3.56a}$$

Repetition of the procedure for $\hat{\boldsymbol{\phi}}$ leads to

$$\hat{\boldsymbol{\phi}} = -\hat{\mathbf{x}}\sin\phi + \hat{\mathbf{y}}\cos\phi. \tag{3.56b}$$

The third base vector $\hat{\mathbf{z}}$ is the same in both coordinate systems. By solving Eqs. (3.56a) and (3.56b) simultaneously for $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$, we obtain the following inverse relations:

$$\hat{\mathbf{x}} = \hat{\mathbf{r}}\cos\phi - \hat{\boldsymbol{\phi}}\sin\phi, \tag{3.57a}$$
$$\hat{\mathbf{y}} = \hat{\mathbf{r}}\sin\phi + \hat{\boldsymbol{\phi}}\cos\phi. \tag{3.57b}$$

The relations given by Eqs. (3.56a) to (3.57b) are not only useful for transforming the base vectors $(\hat{\mathbf{x}},\hat{\mathbf{y}})$ into $(\hat{\mathbf{r}},\hat{\boldsymbol{\phi}})$, and vice versa, they also can be used to transform the components of a vector expressed in either coordinate system into its corresponding components expressed in the other system. For example, a vector $\mathbf{A} = \hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z$ in Cartesian coordinates can be described by $\mathbf{A} = \hat{\mathbf{r}}A_r + \hat{\boldsymbol{\phi}}A_\phi + \hat{\mathbf{z}}A_z$ in cylindrical coordinates by applying Eqs. (3.56a) and (3.56b). That is,

$$A_r = A_x\cos\phi + A_y\sin\phi, \tag{3.58a}$$
$$A_\phi = -A_x\sin\phi + A_y\cos\phi, \tag{3.58b}$$

and, conversely,

$$A_x = A_r\cos\phi - A_\phi\sin\phi, \tag{3.59a}$$
$$A_y = A_r\sin\phi + A_\phi\cos\phi. \tag{3.59b}$$

The transformation relations given in this and the following two subsections are summarized in **Table 3-2**.

Example 3-8: Cartesian to Cylindrical Transformations

Given point $P_1 = (3,-4,3)$ and vector $\mathbf{A} = \hat{\mathbf{x}}2 - \hat{\mathbf{y}}3 + \hat{\mathbf{z}}4$ defined in Cartesian coordinates, express P_1 and $\mathbf{A}$ in cylindrical coordinates and evaluate $\mathbf{A}$ at P_1.

Solution: For point P_1, $x = 3$, $y = -4$, and $z = 3$. Using Eq. (3.51), we have

$$r = \sqrt[+]{x^2+y^2} = 5, \quad \phi = \tan^{-1}\frac{y}{x} = -53.1^\circ = 306.9^\circ,$$

and z remains unchanged. Hence, $P_1 = P_1(5, 306.9^\circ, 3)$ in cylindrical coordinates.

The cylindrical components of vector $\mathbf{A} = \hat{\mathbf{r}}A_r + \hat{\boldsymbol{\phi}}A_\phi + \hat{\mathbf{z}}A_z$ can be determined by applying Eqs. (3.58a) and (3.58b):

$$A_r = A_x\cos\phi + A_y\sin\phi = 2\cos\phi - 3\sin\phi,$$
$$A_\phi = -A_x\sin\phi + A_y\cos\phi = -2\sin\phi - 3\cos\phi,$$
$$A_z = 4.$$

Hence,

$$\mathbf{A} = \hat{\mathbf{r}}(2\cos\phi - 3\sin\phi) - \hat{\boldsymbol{\phi}}(2\sin\phi + 3\cos\phi) + \hat{\mathbf{z}}4.$$

At point P, $\phi = 306.9^\circ$, which gives

$$\mathbf{A} = \hat{\mathbf{r}}\,3.60 - \hat{\boldsymbol{\phi}}\,0.20 + \hat{\mathbf{z}}\,4.$$

3-3.2 Cartesian to Spherical Transformations

From **Fig. 3-18**, we obtain the following relations between the Cartesian coordinates (x,y,z) and the spherical coordinates (R,θ,ϕ):

$$R = \sqrt[+]{x^2+y^2+z^2}, \tag{3.60a}$$
$$\theta = \tan^{-1}\left[\frac{\sqrt[+]{x^2+y^2}}{z}\right], \tag{3.60b}$$
$$\phi = \tan^{-1}\left(\frac{y}{x}\right). \tag{3.60c}$$

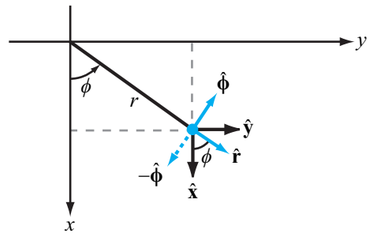


图 3-17 基向量 (x̂, ŷ) 和 (r̂, $\hat{\boldsymbol{\phi}}$) 之间的相互关系。

等式的比较(3.55) 与等式。 (3.53a) 产生 = cos ϕ。类似地，将点积 r̂ · ŷ 应用于等式： (3.54) 给出 b = sin ϕ。因此，

$$\hat{\mathbf{r}} = \hat{\mathbf{x}}\cos\phi + \hat{\mathbf{y}}\sin\phi. \tag{3.56a}$$

重复 $\hat{\boldsymbol{\phi}}$ 的过程会导致

$$\hat{\boldsymbol{\phi}} = -\hat{\mathbf{x}}\sin\phi + \hat{\mathbf{y}}\cos\phi. \tag{3.56b}$$

第三基向量ẑ在两个坐标系中是相同的。通过求解方程。(3.56a)和(3.56b)同时对于x̂和ŷ，我们得到以下逆关系：

$$\hat{\mathbf{x}} = \hat{\mathbf{r}}\cos\phi - \hat{\boldsymbol{\phi}}\sin\phi, \tag{3.57a}$$
$$\hat{\mathbf{y}} = \hat{\mathbf{r}}\sin\phi + \hat{\boldsymbol{\phi}}\cos\phi. \tag{3.57b}$$

等式给出的关系。 (3.56a) 至 (3.57b) 不仅可用于将基向量 (x̂, ŷ) 转换为 (r̂, $\hat{\boldsymbol{\phi}}$)，反之亦然，它们还可用于将在任一坐标系中表达的向量分量转换为在另一坐标系中表达的相应分量。例如，笛卡尔坐标中的向量 A = x̂A_x + ŷA_y + ẑA_z 可以通过应用等式 1 来由柱坐标中的 A = r̂A_r + $\hat{\boldsymbol{\phi}}$A$_\phi$ + ẑA_z 来描述。 (3.56a) 和 (3.56b) 。那是，

$$A_r = A_x\cos\phi + A_y\sin\phi, \tag{3.58a}$$
$$A_\phi = -A_x\sin\phi + A_y\cos\phi, \tag{3.58b}$$

反之，

$$A_x = A_r\cos\phi - A_\phi\sin\phi, \tag{3.59a}$$
$$A_y = A_r\sin\phi + A_\phi\cos\phi. \tag{3.59b}$$

表 3-2 总结了本节和以下两小节中给出的变换关系。

示例 3-8：笛卡尔坐标到圆柱坐标变换

给定笛卡尔坐标中定义的点 P$_1$ = (3、−4、3) 和向量 A = x̂2 − ŷ3 + ẑ4，用柱坐标表示 P$_1$ 和 A，并在 P$_1$ 处计算 A。

解：对于点 P$_1$、x = 3、y = −4 和 z = 3。 (3.51) ，我们有

$$r = \sqrt[+]{x^2+y^2} = 5, \quad \phi = \tan^{-1}\frac{y}{x} = -53.1^\circ = 306.9^\circ,$$

且 z 保持不变。因此，柱坐标系中的 P$_1$ = P$_1$(5, 306.9°, 3) 。

矢量 A = r̂A$_r$ + $\hat{\boldsymbol{\phi}}$A$_\phi$ + ẑA_z 的圆柱分量可以通过应用等式来确定。 (3.58a) 和 (3.58b) ：

$$A_r = A_x\cos\phi + A_y\sin\phi = 2\cos\phi - 3\sin\phi,$$
$$A_\phi = -A_x\sin\phi + A_y\cos\phi = -2\sin\phi - 3\cos\phi,$$
$$A_z = 4.$$

因此，

$$\mathbf{A} = \hat{\mathbf{r}}(2\cos\phi - 3\sin\phi) - \hat{\boldsymbol{\phi}}(2\sin\phi + 3\cos\phi) + \hat{\mathbf{z}}4.$$

在点 P，ϕ = 306.9°，这给出

$$\mathbf{A} = \hat{\mathbf{r}}\,3.60 - \hat{\boldsymbol{\phi}}\,0.20 + \hat{\mathbf{z}}\,4.$$

3-3.2 笛卡尔到球面的变换

从图3-18，我们得到笛卡尔坐标（x，y，z）和球坐标（R，θ，ϕ）之间的关系：

$$R = \sqrt[+]{x^2+y^2+z^2}, \tag{3.60a}$$
$$\theta = \tan^{-1}\left[\frac{\sqrt[+]{x^2+y^2}}{z}\right], \tag{3.60b}$$
$$\phi = \tan^{-1}\left(\frac{y}{x}\right). \tag{3.60c}$$

Table 3-2 Coordinate transformation relations.

Transformation	Coordinate Variables	Unit Vectors	Vector Components
Cartesian to cylindrical	$r = \sqrt[+]{x^2+y^2}$ $\phi = \tan^{-1}(y/x)$ $z = z$	$\hat{\mathbf{r}} = \hat{\mathbf{x}}\cos\phi + \hat{\mathbf{y}}\sin\phi$ $\hat{\boldsymbol{\phi}} = -\hat{\mathbf{x}}\sin\phi + \hat{\mathbf{y}}\cos\phi$ $\hat{\mathbf{z}} = \hat{\mathbf{z}}$	$A_r = A_x\cos\phi + A_y\sin\phi$ $A_\phi = -A_x\sin\phi + A_y\cos\phi$ $A_z = A_z$
Cylindrical to Cartesian	$x = r\cos\phi$ $y = r\sin\phi$ $z = z$	$\hat{\mathbf{x}} = \hat{\mathbf{r}}\cos\phi - \hat{\boldsymbol{\phi}}\sin\phi$ $\hat{\mathbf{y}} = \hat{\mathbf{r}}\sin\phi + \hat{\boldsymbol{\phi}}\cos\phi$ $\hat{\mathbf{z}} = \hat{\mathbf{z}}$	$A_x = A_r\cos\phi - A_\phi\sin\phi$ $A_y = A_r\sin\phi + A_\phi\cos\phi$ $A_z = A_z$
Cartesian to spherical	$R = \sqrt[+]{x^2+y^2+z^2}$ $\theta = \tan^{-1}[\sqrt[+]{x^2+y^2}/z]$ $\phi = \tan^{-1}(y/x)$	$\hat{\mathbf{R}} = \hat{\mathbf{x}}\sin\theta\cos\phi + \hat{\mathbf{y}}\sin\theta\sin\phi + \hat{\mathbf{z}}\cos\theta$ $\hat{\boldsymbol{\theta}} = \hat{\mathbf{x}}\cos\theta\cos\phi + \hat{\mathbf{y}}\cos\theta\sin\phi - \hat{\mathbf{z}}\sin\theta$ $\hat{\boldsymbol{\phi}} = -\hat{\mathbf{x}}\sin\phi + \hat{\mathbf{y}}\cos\phi$	$A_R = A_x\sin\theta\cos\phi + A_y\sin\theta\sin\phi + A_z\cos\theta$ $A_\theta = A_x\cos\theta\cos\phi + A_y\cos\theta\sin\phi - A_z\sin\theta$ $A_\phi = -A_x\sin\phi + A_y\cos\phi$
Spherical to Cartesian	$x = R\sin\theta\cos\phi$ $y = R\sin\theta\sin\phi$ $z = R\cos\theta$	$\hat{\mathbf{x}} = \hat{\mathbf{R}}\sin\theta\cos\phi + \hat{\boldsymbol{\theta}}\cos\theta\cos\phi - \hat{\boldsymbol{\phi}}\sin\phi$ $\hat{\mathbf{y}} = \hat{\mathbf{R}}\sin\theta\sin\phi + \hat{\boldsymbol{\theta}}\cos\theta\sin\phi + \hat{\boldsymbol{\phi}}\cos\phi$ $\hat{\mathbf{z}} = \hat{\mathbf{R}}\cos\theta - \hat{\boldsymbol{\theta}}\sin\theta$	$A_x = A_R\sin\theta\cos\phi + A_\theta\cos\theta\cos\phi - A_\phi\sin\phi$ $A_y = A_R\sin\theta\sin\phi + A_\theta\cos\theta\sin\phi + A_\phi\cos\phi$ $A_z = A_R\cos\theta - A_\theta\sin\theta$
Cylindrical to spherical	$R = \sqrt[+]{r^2+z^2}$ $\theta = \tan^{-1}(r/z)$ $\phi = \phi$	$\hat{\mathbf{R}} = \hat{\mathbf{r}}\sin\theta + \hat{\mathbf{z}}\cos\theta$ $\hat{\boldsymbol{\theta}} = \hat{\mathbf{r}}\cos\theta - \hat{\mathbf{z}}\sin\theta$ $\hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\phi}}$	$A_R = A_r\sin\theta + A_z\cos\theta$ $A_\theta = A_r\cos\theta - A_z\sin\theta$ $A_\phi = A_\phi$
Spherical to cylindrical	$r = R\sin\theta$ $\phi = \phi$ $z = R\cos\theta$	$\hat{\mathbf{r}} = \hat{\mathbf{R}}\sin\theta + \hat{\boldsymbol{\theta}}\cos\theta$ $\hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\phi}}$ $\hat{\mathbf{z}} = \hat{\mathbf{R}}\cos\theta - \hat{\boldsymbol{\theta}}\sin\theta$	$A_r = A_R\sin\theta + A_\theta\cos\theta$ $A_\phi = A_\phi$ $A_z = A_R\cos\theta - A_\theta\sin\theta$

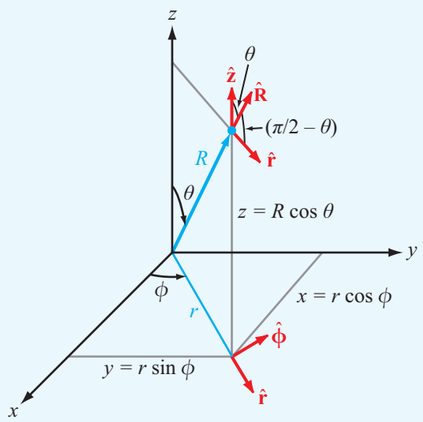


Figure 3-18 Interrelationships between (x,y,z) and (R,θ,ϕ).

The converse relations are

$$x = R\sin\theta\cos\phi, \tag{3.61a}$$
$$y = R\sin\theta\sin\phi, \tag{3.61b}$$
$$z = R\cos\theta. \tag{3.61c}$$

The unit vector $\hat{\mathbf{R}}$ lies in the $\hat{\mathbf{r}}$–$\hat{\mathbf{z}}$ plane. Hence, it can be expressed as a linear combination of $\hat{\mathbf{r}}$ and $\hat{\mathbf{z}}$ as

$$\hat{\mathbf{R}} = \hat{\mathbf{r}}a + \hat{\mathbf{z}}b, \tag{3.62}$$

where a and b are transformation coefficients. Since $\hat{\mathbf{r}}$ and $\hat{\mathbf{z}}$ are mutually perpendicular,

$$\hat{\mathbf{R}}\cdot\hat{\mathbf{r}} = a, \tag{3.63a}$$
$$\hat{\mathbf{R}}\cdot\hat{\mathbf{z}} = b. \tag{3.63b}$$

From Fig. 3-18, the angle between $\hat{\mathbf{R}}$ and $\hat{\mathbf{r}}$ is the complement of θ and that between $\hat{\mathbf{R}}$ and $\hat{\mathbf{z}}$ is θ. Hence, $a = \hat{\mathbf{R}}\cdot\hat{\mathbf{r}} = \sin\theta$

表3-2 坐标变换关系。

Transformation	Coordinate Variables	Unit Vectors	Vector Components
Cartesian to cylindrical	$r = \sqrt[+]{x^2+y^2}$ $\phi = \tan^{-1}(y/x)$ $z = z$	$\hat{\mathbf{r}} = \hat{\mathbf{x}}\cos\phi + \hat{\mathbf{y}}\sin\phi$ $\hat{\boldsymbol{\phi}} = -\hat{\mathbf{x}}\sin\phi + \hat{\mathbf{y}}\cos\phi$ $\hat{\mathbf{z}} = \hat{\mathbf{z}}$	$A_r = A_x\cos\phi + A_y\sin\phi$ $A_\phi = -A_x\sin\phi + A_y\cos\phi$ $A_z = A_z$
Cylindrical to Cartesian	$x = r\cos\phi$ $y = r\sin\phi$ $z = z$	$\hat{\mathbf{x}} = \hat{\mathbf{r}}\cos\phi - \hat{\boldsymbol{\phi}}\sin\phi$ $\hat{\mathbf{y}} = \hat{\mathbf{r}}\sin\phi + \hat{\boldsymbol{\phi}}\cos\phi$ $\hat{\mathbf{z}} = \hat{\mathbf{z}}$	$A_x = A_r\cos\phi - A_\phi\sin\phi$ $A_y = A_r\sin\phi + A_\phi\cos\phi$ $A_z = A_z$
Cartesian to spherical	$R = \sqrt[+]{x^2+y^2+z^2}$ $\theta = \tan^{-1}[\sqrt[+]{x^2+y^2}/z]$ $\phi = \tan^{-1}(y/x)$	$\hat{\mathbf{R}} = \hat{\mathbf{x}}\sin\theta\cos\phi + \hat{\mathbf{y}}\sin\theta\sin\phi + \hat{\mathbf{z}}\cos\theta$ $\hat{\boldsymbol{\theta}} = \hat{\mathbf{x}}\cos\theta\cos\phi + \hat{\mathbf{y}}\cos\theta\sin\phi - \hat{\mathbf{z}}\sin\theta$ $\hat{\boldsymbol{\phi}} = -\hat{\mathbf{x}}\sin\phi + \hat{\mathbf{y}}\cos\phi$	$A_R = A_x\sin\theta\cos\phi + A_y\sin\theta\sin\phi + A_z\cos\theta$ $A_\theta = A_x\cos\theta\cos\phi + A_y\cos\theta\sin\phi - A_z\sin\theta$ $A_\phi = -A_x\sin\phi + A_y\cos\phi$
Spherical to Cartesian	$x = R\sin\theta\cos\phi$ $y = R\sin\theta\sin\phi$ $z = R\cos\theta$	$\hat{\mathbf{x}} = \hat{\mathbf{R}}\sin\theta\cos\phi + \hat{\boldsymbol{\theta}}\cos\theta\cos\phi - \hat{\boldsymbol{\phi}}\sin\phi$ $\hat{\mathbf{y}} = \hat{\mathbf{R}}\sin\theta\sin\phi + \hat{\boldsymbol{\theta}}\cos\theta\sin\phi + \hat{\boldsymbol{\phi}}\cos\phi$ $\hat{\mathbf{z}} = \hat{\mathbf{R}}\cos\theta - \hat{\boldsymbol{\theta}}\sin\theta$	$A_x = A_R\sin\theta\cos\phi + A_\theta\cos\theta\cos\phi - A_\phi\sin\phi$ $A_y = A_R\sin\theta\sin\phi + A_\theta\cos\theta\sin\phi + A_\phi\cos\phi$ $A_z = A_R\cos\theta - A_\theta\sin\theta$
Cylindrical to spherical	$R = \sqrt[+]{r^2+z^2}$ $\theta = \tan^{-1}(r/z)$ $\phi = \phi$	$\hat{\mathbf{R}} = \hat{\mathbf{r}}\sin\theta + \hat{\mathbf{z}}\cos\theta$ $\hat{\boldsymbol{\theta}} = \hat{\mathbf{r}}\cos\theta - \hat{\mathbf{z}}\sin\theta$ $\hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\phi}}$	$A_R = A_r\sin\theta + A_z\cos\theta$ $A_\theta = A_r\cos\theta - A_z\sin\theta$ $A_\phi = A_\phi$
Spherical to cylindrical	$r = R\sin\theta$ $\phi = \phi$ $z = R\cos\theta$	$\hat{\mathbf{r}} = \hat{\mathbf{R}}\sin\theta + \hat{\boldsymbol{\theta}}\cos\theta$ $\hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\phi}}$ $\hat{\mathbf{z}} = \hat{\mathbf{R}}\cos\theta - \hat{\boldsymbol{\theta}}\sin\theta$	$A_r = A_R\sin\theta + A_\theta\cos\theta$ $A_\phi = A_\phi$ $A_z = A_R\cos\theta - A_\theta\sin\theta$

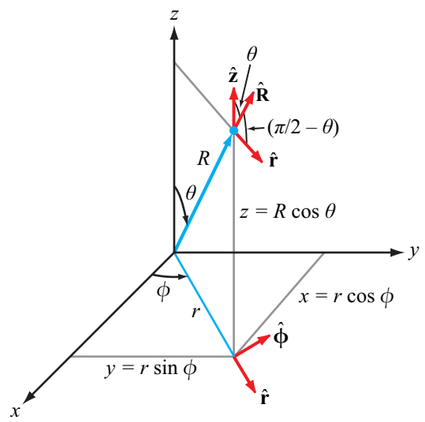


图 3-18 (x,y,z) 和 (R,θ,ϕ) 之间的相互关系。

其逆关系为

$$x = R\sin\theta\cos\phi, \tag{3.61a}$$
$$y = R\sin\theta\sin\phi, \tag{3.61b}$$
$$z = R\cos\theta. \tag{3.61c}$$

单位向量 R̂ 位于 r̂–ẑ 平面。因此，它可以表示为 r̂ 和 ẑ 的线性组合：

$$\hat{\mathbf{R}} = \hat{\mathbf{r}}a + \hat{\mathbf{z}}b, \tag{3.62}$$

其中 a 和 b 是变换系数。由于 r̂ 和 ẑ 相互垂直，

$$\hat{\mathbf{R}}\cdot\hat{\mathbf{r}} = a, \tag{3.63a}$$
$$\hat{\mathbf{R}}\cdot\hat{\mathbf{z}} = b. \tag{3.63b}$$

从图3-18可知，R̂和r̂之间的角度是θ的补角，R̂和ẑ之间的角度是θ。因此，a = R̂ · r̂ = sin θ